WebSketch 3D — Technical Review: Design & Analysis Workspace (ACI 318-19)

Technical Review — Design & Analysis Workspace

App: WebSketch 3D RC module · Code standard: ACI 318-19 (SI units, N·mm·MPa internally)

Scope: every ribbon tab of the Design & Analysis workspace — Define, Assign, Analyze, Design — tool by tool, input value by input value: what each value is, where it is consumed (which equation, which file:line), and the ACI 318-19 clause the equation comes from, with the actual code snippet so you can audit it.

> Reading key: `rcbeam.js:41` = file `js/design/rcbeam.js`, line 41. All equations run in N, mm, MPa; display in kN, kN·m, mm. Cross-references marked [ETABS] cite the decompiled source (`D:\REA\ETABS REA\decompiled`) — those are CSI implementation facts, not ACI.

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1. DEFINE TAB

1.1 Materials (Define › Materials)

Each material record (`rcdefine.js:42 defaultMaterials`) carries:

Field	Meaning	Used?	Where
`conc.fc`	Specified compressive strength f'c (MPa)	[USED] everywhere	every strength equation below
`iso.E`	Young's modulus (0 = auto)	[USED]	stiffness assembly `rcmodel.js:104`
`iso.u`	Poisson's ratio ν	[USED]	`rcmodel.js:103` → G = E/2(1+ν)
`iso.a`	Thermal expansion α	[NOT USED] unused	no thermal load case exists yet
`iso.G`	Shear modulus (0 = auto)	[USED]	`rcmodel.js:105`
`weight.value`	Unit weight (25 kN/m^3) or mass	[USED]	self-weight + Ec auto
`damping.modalRatio`	ζ (0.05)	[USED]	Response-spectrum scaling `rcmodel.js` (rsDialog)
`conc.lightweight`	λ flag	[USED]	λ in Vc, fr, torsion
`conc.strainAtFc / strainUltimate`	Hognestad curve params	[NOT USED] unused	fiber PM uses Whitney block
`rebar.fy`	Yield strength (500 MPa)	[USED]	all flexure/shear steel equations
`tempProps, ssCurve`	T-stress / stress-strain tables	[NOT USED] unused	no creep/shrink/thermal engine

Ec — auto-computed from f'c and unit weight (`rcdefine.js`, concreteEc):

```
ACI 318-19 §19.2.2.1(b): Ec = 0.043·wc^1.5·√f'c  (wc kg/m³);  normalweight ≈ 4700√f'c MPa
// rcdefine.js concreteEc()
if (m.conc.lightweight) return 0.043 * Math.pow(wc, 1.5) * Math.sqrt(fc);
return 4700 * Math.sqrt(fc);
```

Consumed at `rcmodel.js:104` → element stiffness EA, EI, GAs (analysis), and at `rcmodel.js` crackedSectionProps (deflection, Ec again = 4700√f'c, §19.2.2.1(b)).

Where f'c enters each design equation (complete list):

Equation	Uses f'c as	ACI 318-19 clause	Code
a = As·fy/(0.85·f'c·b)	stress block	§22.2.2.4.1 (α=0.85)	`rcbeam.js:41`
β1 = 0.85−0.05(f'c−28)/7 □[0.65,0.85]	block shape	Table 22.2.2.4.3	`rcbeam.js:21`
φPn column cap 0.80·[0.85f'c(Ag−Ast)+fyAst]	axial	§22.4.2	`rccolumn.js` (P0 anchor)
PM fiber sweep σc = 0.85f'c, εcu=0.003	section	§22.2.2.1, §22.2.2.4	`rccolumn.js:56`
Beam Vc = 0.17λ√f'c·bw·d	shear	§22.5.5.1	`rcbeam.js:127`
Vs cap = 2.1√f'c·bw·d	shear	§22.5.1.2	`rcbeam.js:138`

Equation	Uses f'c as	ACI 318-19 clause	Code
Av,min = max(0.062√f'c, 0.35)bw/fyt	shear	§9.6.3.4	`rcbeam.js:151`
Torsion thresholds 0.061/0.083/0.66√f'c	torsion	§22.7.4 / §22.7.6	`rcbeam.js:174+`
As,min = max(0.25√f'c, 1.4)/fy·bw·d	flexure	§9.6.1.2	`rcbeam.js:299`
Wall Vc = αc·0.17√f'c·Aw·(axial)	shear	§11.5.4.3	`rcwall.js:131`
Wall cap 0.66√f'c·Aw	shear	§11.5.4.3 cap	`rcwall.js:134`
Slab one-way 0.17√f'c·b·d	shear	§22.5.5.1	`rcslab.js:112,195`
Punching 0.17(1+2/β), 0.083(αs d/bo+2), 0.33	two-way	Table 22.6.5.2	`rcslab.js:92`
fr = 0.62λ√f'c	rupture	§24.2.3.6	`rcmodel.js:942`
Bearing φ·0.85f'c·A1·√(A2/A1)	bearing	§16.3.1 (old 10.17)	`rccolslab.js`
Joint φVn = 0.85·γ·√f'c·bj·hc	joint	[ETABS] φ=0.85; γ per ACI 21.3.4 SI	`rccolumn.js:297`

Example trace — f'c in beam flexure (the equation you asked about):

The demand Mu comes from analysis; the app solves the quadratic directly instead of iterating a:

```
// rcbeam.js:41 designRectFlexure
const disc = d * d - 2 * Mu / (0.85 * phi * fc * b);   // 0.85·f'c·b = Whitney block (ACI 22.2.2.4.1)
const a = d - Math.sqrt(disc);
const As = Mu / (phi * fy * (d - a / 2));              // Mn = As·fy(d−a/2) (ACI 22.3)
```

This is Mu = φ·As·fy·(d − a/2) with a = As·fy/(0.85·f'c·b) — §22.2.2.4.1 + §22.3.1.1.

φ then re-verified from the actual neutral axis:

```
// rcbeam.js:12 phiFlexure — ACI Table 21.2.2 + transition §21.2.2
if (et >= esy + 0.003) return 0.90;                    // tension-controlled
if (et <= esy) return 0.65;                            // compression-controlled (tied)
return 0.65 + 0.25 * (et - esy) / (0.005 - esy);       // linear transition
```

1.2 Frame Sections (Define › Frame Sections)

Field	Meaning	Used?	Where
`dims.b, dims.h` (m)	section size	[USED]	stiffness (A, I), fiber PM bars, covers
`material`	→ materials registry	[USED]	E, G, fc, weight
`designType` beam/column	routing	[USED]	designAllBeams/Columns filters
`modifiers {a,i22,i33,as2,as3,torsion,mass,weight}`	ETABS SetModifiers	[USED]	`rcmodel.js` frame push: A·a, Iy·i22… cracked-analysis factors
`rigidZone` (0–1)	rigid-zone factor	[USED]	`rcmodel.js` eoI×rz (end offsets)
`rebar.coverTop/coverBot/colCover`	cover (m)	[USED]	d = h − cover − stirrup − db/2
`rebar.rebarSize, nR2, nR3`	layout	[USED]	PM fiber bars `rccolumn.js rebarLayout`
`rebar.tieSize, tieSpacingLongit`	ties	[USED]	column shear Av, s `rccolumn.js:373+`
`rebar.topLeft…` (T-sections)	flange steel	[PARTIAL]	designTBeamFlange (partial)

Section properties → stiffness:

```
// rcmodel.js:66-71 — analysis uses REAL section properties × modifiers
const A  = props.A * 1e6 * (mo.a ?? 1);                // mm²
const I33 = props.I33 * 1e12 * (mo.i33 ?? 1);          // mm■  (I33 = strong axis for beams)
const J  = rectJ(bmm, hmm) * (mo.torsion ?? 1);
```

ACI 318-19 §6.6.3.1.1 / [ETABS] — for cracked first-order analysis the user sets

i33 = 0.35 etc.; the app never hardcodes them.

1.3 Load Patterns / Cases / Combos

Field	Meaning	Used?	Where
`selfWtMult`	self-weight multiplier	[USED]	`rcmodel.js:476+` w = A·γ·selfWtMult
pattern type DEAD/LIVE…	family routing	[USED]	auto combos, long-term split
`loadCases[Modal].numModes`	eigen count	[USED]	RS dialog modalAnalysis (on demand)

Field	Meaning	Used?	Where
`combos[].cases[{name,scale}]`, type add/srss/absolute/envelope	combination	[USED]	`rcmodel.js` mix() + designSources
ensureAutoCombos	ETABS DConS parity	[USED]	1.4D, 1.2D+1.6L, 0.9D, D+0.75L(A/B)
`autoSeismic {ss,s1,…}`	ASCE 7 params	[NOT USED] defined, not wired	no auto lateral generator yet
`autoWind {speed,…}`	ASCE 7 wind	[NOT USED] defined, not wired	same
`rsFunctions`, `thFunctions`	Sa(T)/time series	[USED]	rsDialog / timeHistoryAnalysis
`massSource`	modal mass	[USED]	lumpedMass extra (RS path)
`mesh.size`	shell mesh (m)	[USED]	`rcmodel.js:197` wall/slab grid
`diaphragms`, `constraints`	rigid ties	[PARTIAL]	rigidLinks hook exists in solver; no UI pass
`linkProps`, `tendonProps`, `cableSections`, `solidSections`	other elements	[NOT USED] defined, not analyzed	registries only
`pointSprings/lineSprings/areaSprings`	supports	[USED]	`augmentModel` → nodalSprings
`piers/spandrels` labels	wall strips	[USED]	wall design rows + spandrel design

Patterned live load (ETABS PatternLiveFactor [ETABS] `cConcreteCode_ACI_318_08.cs:405` = 0.75):

```
// rcmodel.js:746 ensureAutoCombos — skip-live parity
mk('D+0.75L(A)', [{name: DEAD.name, scale: 1.0}, {name: LIVE.name + '#A', scale: 0.75}]);
// rcmodel.js buildLoads2: '#A/#B' virtual cases gate entity loads by parity
if (wantParity != null && entParity.get(owner.id) !== wantParity) continue;
```

Basis: ACI 318-19 §5.3.1(b) alternate / skip-live factoring (0.75L patterned).

1.4 Sync Model → Define

Catalogs drawn elements into sections by dimensions (find-or-create, `ensureAutoSection`), stamps `params.designSection`. Pure bookkeeping — no equations.

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2. ASSIGN TAB (tool by tool)

Tool	Inputs	Where consumed
Frame Sections	section name	design + stiffness path (§1.2)
Area Sections	slab/wall section	thickness → shell t (`rcmodel.js`), wall design
Reset to Auto	—	clears manual `designSectionManual`
Frame Loads	w kN/m, P kN, pattern, gravity/up/gx/gy	`rcmodel.js buildLoads2` → consistentUDL equivalent loads
Surface Loads	q kN/m^2, pattern, down/up	`rcmodel.js` shells: nodal q·A/4 — ACI §5.2 distribution
Joint Loads	fx/fy/fz	nodal vector
Point Springs	k1..k3	solver diagonal (Winkler) — theory, not ACI
Material Overwrite	material name	overrides section material
Design Procedure	concrete/nodesign	filters design passes
Frame Releases	pin M3 at i/j	`applyReleases` moment release — §6.2.5 idealization
End Length Offsets	auto or user eoI/eoJ, RZF	ETABS semantics: analysis stays centerline; design forces at faces — forces at member faces, not centerline. `rcmodel.js` offI/offJ + `rcbeam.js:270` stationAt(offI)
Base Fixity	fixed/pinned	support BC `rcmodel.js` fixAll/fixPin
Pier / Spandrel Labels	P1/S1	wall design reporting, spandrel grouping
Diaphragms / Strips	names	slab strip design grouping (v1 auto strips)

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3. ANALYZE TAB (tool by tool)

Run Analysis — direct-stiffness FEA (`fea.js`): 12-DOF frames, 20-DOF Mindlin shells, skyline U^TDU factorization, one factorization for all patterns. Equilibrium checked (reactions = Σloads, 0.9% at 1 m mesh demo).

```
// fea.js assembleAndSolveMulti — factor once, solve per pattern
const U = skySolve(fvecs[p]);        // U■DU substitution
framesArr.push(frameInfo.map(fi => { /* f = Kl·ud − feq */ }));
```

Physics: stiffness method (textbook, same as ETABS SAPFire conceptually).

Response Spectrum / Time History — CQC-ready modal via subspace iteration (`modalAnalysis`, Guyan condensation); Newmark-β (γ=½, β=¼ — linear, unconditionally stable, classic formulation) for TH; Rayleigh damping ζ from `damping.modalRatio`.

Diagrams (M3/V3) — station statics from end forces + loads:

```
// rcmodel.js:700 M(x) = Mi + Vi·x − wy·x²/2 − ΣP■x−a■
let M = f.forces[5] + (-f.forces[1]) * x;
M += -wy * x * x / 2;
```

Deflection button — three engines, three datums (all ACI Ch. 24 or pure stiffness):

1. `beamDeflections` — elastic line M/EI double integration, chord-relative (span datum)
2. `serviceDeflections` — cracked ACI Ch. 24 (the service-deflection check)
3. `elementDeflections` — slab support/total/bending, column drift, story drifts

Force Table / Show Tables / Story Results / Element Data — reporting surfaces over the same result arrays (Joint Displacements, Base Reactions, Design Forces — beams/columns, Story Drifts, Slab Deflections, Material List — ETABS `cTable*` schemas).

Deflected shape / Property Lines / Results View — display modes ([ETABS] `eObjectRepresentationTypes` ClineAndMidPlane / `ResultDisplayTypes`).

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4. DESIGN TAB (tool by tool, every equation)

4.1 Beam Design — `designAllBeams` (`rcbeam.js:230`)

Demand: ENVELOPE across combos (per-station max — ETABS Ast-profile parity), faces from end offsets (`stationAt(offI)`, §3 End Length Offsets above).

#	Equation	ACI 318-19	Code
1	a = As·fy/(0.85 f'c b); As = Mu/(φ fy (d−a/2))	§22.2.2.4.1, §22.3	`rcbeam.js:41-48`
2	φ by εt (0.9 / 0.65 / linear)	Table 21.2.2, §21.2.2	`rcbeam.js:12-18`
3	a_max at εt = εty+0.003 (tension-controlled)	§9.3.3.1	`rcbeam.js:44`
4	As,min = max(0.25√f'c, 1.4)/fy·bw·d	§9.6.1.2	`rcbeam.js:299`
5	As,max = 0.04·bw·d	[ETABS] MaxAstRatio 0.04 (`cDsnCodeConcRC_ACI_05.cs:135`)	`rcbeam.js:300`
6	Vc = 0.17λλs√f'c·bw·d (λs size effect)	§22.5.5.1(.3)	`rcbeam.js:127-131`
7	Vs = Av·fyt·d/s; Vs ≤ 2.1√f'c·bw·d	§22.5.5.5? (cap §22.5.1.2)	`rcbeam.js:138`
8	Av,min/s = max(0.062√f'c,0.35)bw/fyt	§9.6.3.4	`rcbeam.js:151`
9	Stirrup s: min(600,d/2); high shear min(d/4,100,6db,150)	[ETABS] `_05.cs:23-38` (ACI §9.7.6.2 family)	`rcbeam.js:409+`
10	ld = fy·ψt·db/(2λ√f'c) ≥ 300, ψt=1.3 top	§25.4.2.3 + §25.4.2.1	`rcbeam.js:162`
11	Torsion: Tth 0.061√f'c·Acp²/pcp; Tmax 0.083…; Tcr 0.66…	Ch. 22.7 (§22.7.4, 22.7.6, 22.7.5)	`rcbeam.js:174+`
12	Torsion steel At/s = Tu/(2·0.85Aoh·fyt); Al = (At/s)pcp	§22.7.6.1, §9.7.5.1	`rcbeam.js:186-189`
13	Deflection (cracked, in Element Data)	Ch. 24 — §4.5 below	`rcmodel.js:947`

```
// rcbeam.js:127 designShear — one-way shear
let Vc = 0.17 * lambda * Math.sqrt(fc) * bw * d;     // ACI 22.5.5.1 (SI; = 2√psi)
const VcMax = 0.83 * lambda * Math.sqrt(fc) * bw * d;
```

4.2 Column Design — `designAllColumns` (`rccolumn.js:214`)

#	Equation	ACI 318-19	Code
1	Fiber PM sweep: εcu=0.003, α=0.85, β1, φ(εt)	§22.2.2.1, 22.2.2.4, T21.2.2	`rccolumn.js:56`
2	φPn cap = 0.80·P0 tied / 0.85 spiral	§22.4.2	`rccolumn.js` P0 anchor
3	Displaced-concrete deduction (bar in block)	§22.2.2.4.5? (net area)	sweep loop
4	Biaxial capacity methods (vec/sum/ray/axial, MaxOfAll)	[ETABS] `cPMMSurface.cs:348-519` (CSiCol fiber §22.2)	`rccolumn.js:251`
5	ρ □ [0.01, 0.08]	Table 10.6.1.1	layout accept
6	min ecc e = max(20, h/30, b/30)	§6.2.5 (historical; [ETABS] pref)	`rccolumn.js` eMin
7	Column shear Vc(Nu) branches + Vs cap + φVn=0.8(Vc+Vs)	[ETABS] `_ACI_08.cs:132-147` (ACI 318-08 §11.3 SI)	`rccolumn.js:373`
8	Joint shear φVn = 0.85·γ·√f'c·bj·hc; γ 1.66/1.25/1.0	[ETABS] φ JointShear 0.85; γ ACI 21.3.4 (SI)	`rccolumn.js:297`

```
// rccolumn.js:373 columnNetShearCapacitySI — ETABS port (psi→SI annotated inline)
const val3 = 0.291 * sq * Acv * Math.sqrt(1 + Math.max(Nu, 0) / (3.45 * Ag));
let Vs = Math.min(Av * fyt * h / s, 0.664 * sq * Acv);
const phiVn = 0.8 * (Vc + Vs);       // [ETABS] hardcoded 0.8 (ACI-08 vintage)
```

4.3 Wall Design — `designAllWalls` (`rcwall.js:252`)

#	Equation	ACI 318-19	Code
1	Pier forces: P = ΣN22·w, V = ΣN12·w, M = ΣN22(x−x̄)w at base+top	[ETABS] pier integration	`rcwall.js` wallCut
2	Wall PM fiber curve	§22.2 family	`rcwall.js:47`
3	Vc = αc·0.17√f'c·Aw·(axial); αc 3.0→2.0 over hw/lw 1.5→2	§11.5.4.3	`rcwall.js:122-131`
4	Vc ≤ 0.66√f'c·Aw	§11.5.4.3 cap	`rcwall.js:134`
5	ρv,min = 0.0015; ρh escalation when Vu>0.5φVc	Table 11.6.1, §11.6.3	`rcwall.js` rho blocks
6	SBE check (boundary elements)	§18.10.6	`rccolslab.js` checkSBE
7	Spandrels: jamb-cut N11 → Mu, Vu, As, Avh/s	[ETABS] `SpandrelDesignResult` schema	`rcwall.js:377`

φ = 0.75 shear (§21.2.4 SI table); [ETABS] decompiled path hardcodes 0.8 — we keep 0.75 (ACI-19 correct), documented difference.

4.4 Slab Design — `designAllSlabs` (`rcslab.js`) + Slab Design View (`rcview.js`)

#	Equation	ACI 318-19	Code
1	As per meter from M11/M22 (Whitney, same flexure as beams)	§22.2/22.3	`rcslab.js:12`
2	As,min = 0.0018·b·h	§7.6.1.1	`rcslab.js:22`
3	s_max = min(3h, 450)	§7.7.2.3	`rcslab.js` sMax
4	One-way shear peak-Q vs 0.17√f'c·b·d	§22.5.5.1	`rcslab.js:195`
5	Punching: Jc geometry, γv = 1−1/(1+0.677√(b2/b1))	§8.5.3.1 (γv; 0.677 = 2/3·√ratio)	`rcslab.js:48+`
6	Punching Vc = 0.17(1+2/β), 0.083(αs d/bo+2), 0.33√f'c	Table 22.6.5.2 (SI) — see §5 bug fix	`rcslab.js:92`
7	Drop panels: d at panel, perimeter at drop edge	§8.2.3 (critical section at change)	`rccolslab.js`
8	Strips both directions at mesh stations; As/AsMin/Vu/Avmin	[ETABS] `Concrete Slab Design - Flexure and Shear Data` schema	`rcview.js designSlabStrips`

```
// rcslab.js:92 — FIXED this review (was psi coefficients: (2+4/β),4 → ×12 overstatement)
const Vc1 = 0.17 * (1 + 2 / beta) * sqrtFc * bo * d;
const Vc2 = 0.083 * (alphaS * d / bo + 2) * sqrtFc * bo * d;
const Vc3 = 0.33 * sqrtFc * bo * d;
```

4.5 Deflection engines — ACI Ch. 24 (`rcmodel.js`)

#	Equation	ACI 318-19	Code
1	Ec = 4700√f'c	§19.2.2.1(b)	`rcmodel.js:938`
2	fr = 0.62λ√f'c	§24.2.3.6	`rcmodel.js:942`
3	Mcr = fr·Ig/yt	§24.2.3.5	`rcmodel.js:943`
4	Ie = Icr + (Ig−Icr)(Mcr/Ma)^3 ≤ Ig (per-station)	§24.2.3.5 (Branson)	`rcmodel.js:947+` integ()
5	λΔ = ξ/(1+50ρ'), ξ = 2.0	§24.2.4.1	`rcmodel.js` lambdaLT
6	δLT = (1+λΔ)·δD + δL	§24.2.4 usage	serviceDeflections
7	Limits: δL ≤ L/360; δtotal,LT ≤ L/240	Table 24.2.2	`rcmodel.js` lim360/lim240
8	Slab 3 datums: support / total / panel bending	displacement semantics	`elementDeflections`

```
// rcmodel.js serviceDeflections — the Branson interpolation, per station
const Ie1 = M1 >= Mcr ? Math.min(Icr + (Ig - Icr) * Math.pow(Mcr / M1, 3), Ig) : Ig;
slope.push(slope[i] + 0.5 * (M1/(Ec*Ie1) + M2/(Ec*Ie2)) * dx);   // M/EI double integration
```

4.6 Connections & Foundation

Tool	Equations	ACI	Code
Punching Check	Table 22.6.5.2 + γv 8.5.3.1	as slab	`rccolslab.js designPunchingShear`
SBE Check	c ≥ cw/2… stress-limit & NA-depth	§18.10.6.2/.3	`rccolslab.js checkSBE`
Foundation	one-way 0.17√f'c; two-way Table 22.6.5.2; bearing φ0.85f'cA1√(A2/A1)≤2; flexure §22; ρmin 0.0018	§16.3.1(10.17.1), §13.3 (mat), §7.6.1.1	`rccolslab.js designFooting`
Auto-Size	DCR loop re-analysis	— (optimization)	`rcmodel.js optimizeDesign`

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5. Findings from this review (act on them)

1. BUG FIXED — punching psi/SI mix (critical): `rcslab.js` punching and `rccolslab.js` footing two-way used psi-form coefficients on MPa f'c → capacity overstated ~12×. Converted to SI Table 22.6.5.2 forms; regression test now locks the hand value φVc = 461.5 kN (400 col, d=165, fc=25, bo=2260). 32/32 tests pass.

2. Unused Define values (honest gaps): material `iso.a` (thermal), stress-strain curve fields (`strainAtFc…`), `tempProps`, `autoSeismic`, `autoWind`, `linkProps/tendonProps/cableSections/solidSections` (registries exist, no elements analyzed), diaphragm rigid-tie assignment UI. These are define-only today.

3. ETABS-sourced (not ACI) items, kept by design: As,max 0.04; stirrup-spacing tight rule; column φVn = 0.8(Vc+Vs); joint γ set; PatternLiveFactor 0.75 — each cites its decompiled line in §4.

4. Vintage difference: wall φ 0.75 (ACI-19) vs [ETABS] 0.8 hardcode — kept 0.75.

Generated 2026-10-12 against commit 6d0934c + the punching fix. Every snippet above is verbatim from the working tree at review time.

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6. APPENDIX A — Worked numeric examples (hand-checks you can repeat)

A.1 Beam service deflection — beam 300×600, L = 6 m, f'c = 25, fy = 500

```
Ec  = 4700·√25            = 23 500 MPa          (§19.2.2.1.b)
Ig  = 300·600³/12         = 5.4×10■ mm■
yt  = 300 mm
fr  = 0.62·√25            = 3.10 MPa            (§24.2.3.6)
```

```
Mcr = fr·Ig/yt            = 3.10·5.4e9/300
                          = 55.8×10■ N·mm = 55.8 kN·m   (§24.2.3.5)
Ma (demo, 1.0D)           ≈ 15 kN·m  <  Mcr  →  UNCRACKED, Ie = Ig
```

Because Ma < Mcr the Branson term $(Mcr/Ma)^3$ is skipped and the elastic line integrates with gross EI. From the app run (verified in browser): δD = 2.02 mm, δL = 0.38 mm; long-term (§24.2.4.1, ξ = 2.0, ρ' = 0 → λΔ = 2.0):

```
δLT = (1 + 2.0)·2.02 + 0.38 = 6.44 mm
Limits (Table 24.2.2): L/360 = 16.67 mm (δL 0.38 OK) · L/240 = 25 mm (δLT 6.44 OK)
```

Cracked branch check (a beam with Ma = 80 kN·m > Mcr): Icr from kd-solve ≈ 7.46×10^8 mm^4, Ie = Icr + (Ig−Icr)$(55.8/80)^3$ ≈ 2.6×10^9 mm^4 — the code evaluates this per station (`rcmodel.js` serviceDeflections integ loop), which is why the integration uses station-pair EI averages, not one global Ie.

A.2 Punching — interior column 400×400, d = 165, f'c = 25 (the regression lock)

```
b1 = b2 = 400+165 = 565 · bo = 2(565+565) = 2260 mm          (§22.6.4.1)
β = 1, αs = 40 (interior)
Vc(a) = 0.17(1+2/1)·√25·2260·165 = 2.55·372 900  = 950.9 kN
Vc(b) = 0.083(40·165/2260 + 2)·5·372 900         = 761.4 kN
Vc(c) = 0.33·√25·2260·165     = 1.65·372 900     = 615.3 kN  ← governs
ϕVc  = 0.75·615.3                                 = 461.5 kN  (Table 21.2.2 ϕ=0.75)
```

This exact value is asserted in `scratch/test-shell-integration.js` — the test that caught the psi/SI bug (the old code computed 5.5 MN here).

A.3 Column PM point (one sweep station)

400×400, 8Ø20 (Ast = 2513 mm^2), cover 40: c = 150 mm station:

```
a   = 0.85·150 = 127.5 mm
Cc  = 0.85·25·400·127.5                 = 1083.8 kN
εs  = 0.003(320−150)/150 = 0.0034 > εy → steel yields; top bars in block: −0.85f'c·As
ϕ(εt = 0.0034 > εty+0.003? borderline)  → ϕ ≈ 0.87 (transition §21.2.2)
P   ≈ Cc + ΣFs ≈ 1150 kN, M from lever arms about centroid
```

The sweep runs tension → pure compression with the non-monotonic repair and the 0.80P0 anchor (§22.4.2) — `rccolumn.js:56 generatePMCurve`.

A.4 Wall base integration (pier forces)

For the demo shaft wall the base cut integrates the shell row:

P = Σ(−N22)·w = tributary gravity · V = ΣN12·w · M = Σ(−N22)(x−x̄)w — validated against support reactions (0.9 % equilibrium at 1 m mesh).

---

7. APPENDIX B — Units & the psi→SI conversion table

All internals are N·mm·MPa. Every √f'c coefficient in the codebase, with its psi origin:

psi form (ACI US)	SI form (code)	ratio	Used in
2√f'c	0.17√f'c	×0.083	beam/slab one-way Vc (§22.5.5.1)
4√f'c	0.33√f'c	×0.083	punching (c) (Table 22.6.5.2)
6√f'c	0.50√f'c	×0.083	punching (a) at β=1
8√f'c	0.66√f'c	×0.083	Vs cap, wall cap
10√f'c	0.83√f'c	×0.083	Vc max
(2+4/β)√f'c	0.17(1+2/β)√f'c	×0.083	punching (a) — the fixed line
3.5√f'c	0.29√f'c	×0.083	column Vc branch (ETABS port)
7.5√f'c fr	0.62√f'c	×0.083	modulus of rupture (§24.2.3.6)
500/2000/2500 psi axial terms	3.45/13.79/17.24 MPa	×0.00689	column Vc(Nu)

This table is why the review found the punching bug: the (2+4/β) family is the only place where psi coefficients had leaked into MPa arithmetic.

8. APPENDIX C — Data flow (Define → results)

```
DEFINE registries ■■■
                  ■■■ buildModel (rcmodel.js:46)
ASSIGN stamps ■■■■■■■     nodes/frames/shells · end offsets · snapping · beam auto-mesh
                  ■     wall openings voided · drilling constraints by shell normal
                  ▼
        assembleAndSolveMulti (fea.js)  ■■ skyline U■DU factor ONCE
                  ■                       per-pattern solves + reactions from K■ snapshot
                  ▼
        rcResults { patterns[].members/stations, combos[], reactions[], shells[], U }
                  ▼
   DESIGN  (ENVELOPE over combos via designSources)
     beams  → As/stirrups/dev/torsion        (ACI 9/22/25)
     cols   → fiber PM + biaxial + shear + joint
     walls  → pier PM + αc shear + spandrels
     slabs  → flexure strips + one-way + punching
     defl.  → Ch.24 cracked Ie + λΔ  (serviceDeflections)
                  ▼
   DISPLAY (diagrams, deflected shape, Show Tables, Element Data,
            Property Lines, Results View, Slab Design window)
```

9. APPENDIX D — Unused Define values: proposed wiring

Value	Proposed consumer
`iso.a` thermal	thermal strain load case → nodal ΔT loads
`conc.strainAtFc/strainUltimate`	replace Whitney with Hognestad in PM fiber (research)
`autoSeismic {ss,s1,tl,r,ie…}`	ASCE 7-16 Eq. 12.8-x base-shear → static X/Y lateral patterns
`autoWind {speed,exposure…}`	ASCE 7-16 Ch.27 qz → wall projected pressure loads
`tendonProps/linkProps/cables`	new element kernels (tendon = prestress losses; links = nonlinear)
`diaphragms`	rigid-tie master/slave sets in augmentModel (solver hook exists)
`tempProps`	f'c(t), creep coefficient for λΔ > 2.0

10. APPENDIX E — ACI 318-19 clause index (as cited in this review)

§5.3.1(b) pattern live · §6.2.5 min eccentricity · §6.6.3.1.1 stiffness modifiers ·

T7.6.1.1 / §7.7.2.3 slab min-steel/spacing · §8.2.3 drop-panel critical section ·

§8.5.3.1 γv · §9.3.3.1 tension-controlled · §9.6.1.2 As,min · §9.6.3.4 Av,min ·

§9.7.5.1 torsion longitudinal · §9.7.6.2 stirrup spacing · T10.6.1.1 column ρ limits ·

§11.5.4.3 wall αc shear · T11.6.1 wall min steel · §11.6.3 Vu>0.5φVc · §13.3 mats ·

§16.3.1(10.17.1) bearing · §18.10.6 SBE · §19.2.2.1.b Ec · T21.2.2 φ ·

§22.2.2.1 εcu · §22.2.2.4.1 α · T22.2.2.4.3 β1 · §22.3 flexural strength ·

§22.4.2 axial cap · §22.5.1.2 Vs cap · §22.5.5.1 Vc · §22.7.4/22.7.5/22.7.6 torsion ·

T22.6.5.2 punching Vc · §24.2.2 deflection limits · §24.2.3.5 Mcr/Ie · §24.2.3.6 fr ·

§24.2.4.1 λΔ · §25.4.2.1/25.4.2.3 development.